

Configuration

💡 Only bearings 3 & 4 are shown (these are the only bearings that failed in the dataset)

Select Bearing

3

Select Axis

x



Bearing Failure Prediction Dashboard

Real-time monitoring and RUL prediction | 2.88h MAE (critical zone)

Current RMS

0.5914

Current Kurtosis

8.39

Actual RUL

0.0 hours

Predicted RUL

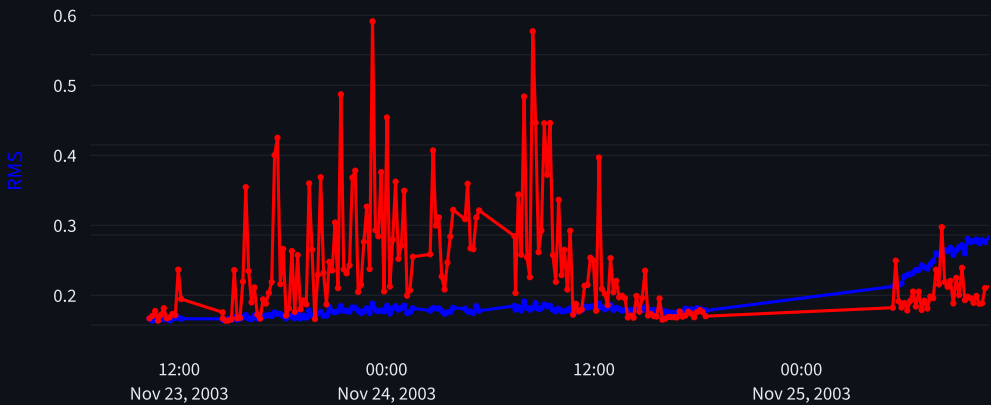
3.5 hours

↑ 3.5h error



CRITICAL: Schedule emergency maintenance immediately!

Degradation Trends (Last 200 Measurements)

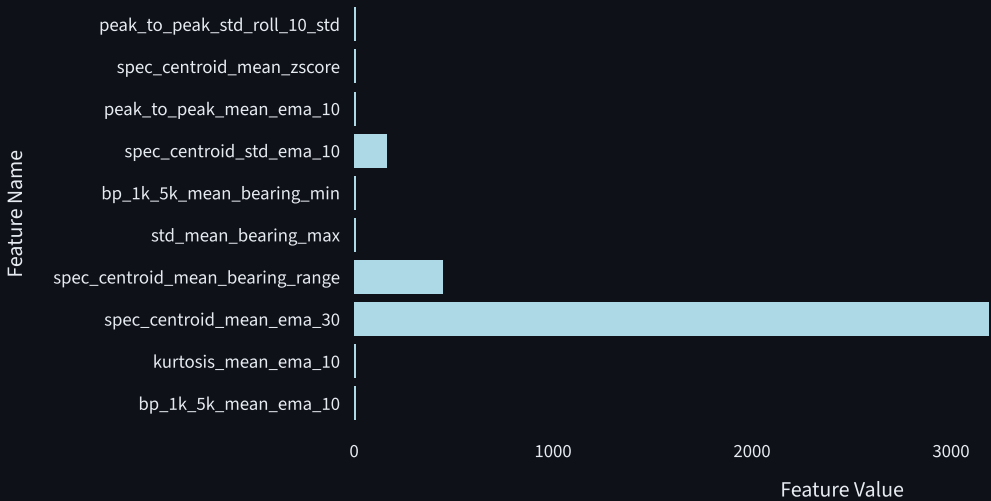


RUL Degradation Over Time



Top 10 Most Important Features (Current Values)

Feature Values at Current Time (Bearing 3-x)



Summary Statistics

Metric	Value	Metric	Value
Mean RMS	0.1956	Mean Kurtosis	6.16
Max RMS	0.5914	Max Kurtosis	33.72
Min RMS	0.1636	Min Kurtosis	0.49
Current RMS	0.5914	Current Kurtosis	8.39

Dataset Information

Total Measurements	Initial RUL	Time to Failure
200	55.9 hours	2.3 days

Model Details:

- Algorithm: LightGBM v2 (Optuna-tuned, 80 trials)
- Critical Zone MAE: 2.88 hours (0-50h RUL range)
- Overall MAE: 13.42 hours
- R²: 0.9852 (98.5% variance explained)
- Dataset: NASA IMS Bearing Dataset
- Features: 50 selected from 380 engineered features
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